

$$f(x, y, z) = x^2 y / z$$

$$\frac{\partial f}{\partial x} = 2xy/z$$

$$\frac{\partial f}{\partial y} = \frac{x^2}{z}$$

$$\frac{\partial f}{\partial z} = -\frac{x^2 y}{z^2}$$

use eq 2

$$df = \sigma_f = \sqrt{\left(\frac{2xy}{z}\right)^2 \sigma_x^2 + \left(\frac{x^2}{z}\right)^2 \sigma_y^2 + \left(\frac{x^2 y}{z^2}\right)^2 \sigma_z^2}$$

$$\frac{\sigma_f}{f} = \sqrt{\left(\frac{2xy}{z}\right)^2 \sigma_x^2 \frac{1}{f^2} + \left(\frac{x^2}{z}\right)^2 \sigma_y^2 \frac{1}{f^2} + \left(\frac{x^2 y}{z^2}\right)^2 \sigma_z^2 \frac{1}{f^2}}$$

$$= \sqrt{\left(\frac{2\sigma_x}{x}\right)^2 + \left(\frac{\sigma_y}{y}\right)^2 + \left(\frac{\sigma_z}{z}\right)^2}$$

the two comes from the power
in $x^2 y / z$.

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$$I = I_0 e^{-\mu x}$$

$$f(I_0, \mu, x)$$

$$\sigma_I \text{ or } \sigma_f = \text{use eq 2}$$

$$\frac{\partial I}{\partial I_0} = e^{-\mu x}$$

$$\frac{\partial I}{\partial \mu} = I_0 (-x) e^{-\mu x}$$

$$\frac{\partial I}{\partial x} = I (-\mu) e^{-\mu x}$$

$$\sigma_I = \sqrt{\left[(e^{-\mu x})^2 \sigma_{I_0}^2 + (I_0 x e^{-\mu x})^2 \sigma_{\mu}^2 + (I)^2 \sigma_x^2 \right]}$$

$$3 \ a) \ C_{max} = \frac{2 + 2(0.005)}{(6.78 \times 10^{-9} - 2.4 \times 10^{-10}) - 2(0.1 \text{E}-9)}$$

$$= 3.180 \times 10^8 \text{ m/s}$$

$$C_{min} = \frac{2 - 2(0.005)}{(\Delta t) + 2(0.1 \text{E}-9)}$$

$$= 2.961 \times 10^8 \text{ m/s}$$

$$\sigma_c = \frac{C_{max} - C_{min}}{2}$$

Very rough

$$= 1.1 \times 10^7 \text{ m/s}$$

$$C = \frac{2}{\Delta t} \pm \sigma_c$$

$$= (3.07 \pm 0.11) \times 10^8 \text{ m/s}$$

Very rough estimate

b) adjustment for N points
- Rough estimate

$$\sigma_{c \text{ rough}} = \frac{\sigma_{c \text{ very rough}}}{\sqrt{N}}$$

$$= 0.03 \times 10^8 \text{ m/s}$$

C

best fit

$$C = (2.99 \pm 0.03) \times 10^8 \text{ s}$$

Note

$\sigma_{C, \text{lin fit}}$ is from stat scatter of data only.

$\sigma_{C, \text{lin fit}}$ agrees well with the rough experimental estimate of uncertainty. - As it should if there are no additional sources of error.

I don't ask if two results for C agree. There is only one best result

$C = 2.99 \dots$ from all the data being used.